

Faculty of Informatics
BCA CBCS-V Semester Regular Examinations November-2025
PAPER: Data Science with R

Time: 3 Hours

Max Marks: 70

Section-A

I. Answer any Five of the following questions (5x4=20 Marks)

1. What is R.
2. Perform any five simple mathematical calculations in R.
3. Write short notes on lists in R with an example.
4. How are date values handled in R? Illustrate with examples.
5. What is the role of the `tapply()` function in R? Explain with syntax and example.
6. Explain how data can be read from the web or APIs into R.
7. List the assumptions of linear regression models.
8. Explain the basic concept of K-means clustering.

Section-B

II. Answer all the following questions (5x10=50 Marks)

9. (a) How can you handle packages in R? Explain with examples.

OR

(b) What are some basic commands used for data exploration in R? Give examples.

10. (a) What are factors in R? How are they created and used? Explain with examples.

OR

(b) How is missing data treated in R? Discuss different methods.

11. (a) How can spreadsheets be read into R? Mention the required packages.

OR

(b) Describe how to load and inspect data frames in R.

12. (a) Write R commands to create a bar chart and pie chart using sample data.

OR

(b) Explain the steps in performing simple linear regression in R.

13. (a) What is the K-Nearest Neighbor (KNN) algorithm? Explain how to implement it in R.

OR

(b) What are association rules? How are they used in data mining? Explain the concept of frequent itemsets with an example.

Faculty of Science
BCA CBCS-V Semester Regular Examinations November-2025
PAPER: Unix Programming

Time: 3 Hours

Max Marks: 70

Section-A

I. Answer any Five of the following questions (5x4=20 Marks)

1. What are the main features of Unix?
2. Write about Unix File System Structure?
3. What is the use of "grep" command?
4. Give one example of using "sed" for text substitution?
5. What is SSH and why it is used?
6. What is Shell Programming?
7. What are looping constructs in shell scripting?
8. What is Inter-Process Communication (IPC)?

Section-B

II. Answer all the following questions (5x10=50 Marks)

9. (a) Explain the architecture of Unix with a neat diagram?
OR
(b) Write about process management and system monitoring commands in Unix?
10. (a) How are system updates and dependency handling managed in Unix/Linux?
OR
(b) Discuss package management in Unix/Linux using "apt" and "yum"?
11. (a) Explain networking tools "ifconfig", "ip", "ping", "netstat" with suitable examples?
OR
(b) Write in detail about Git commands used for collaboration and tracking changes?
12. (a) Write about operators in Shell programming with example?
OR
(b) Discuss the features of text editors "vi", "vim" and "nano"?
13. (a) Explain different IPC Mechanisms and its advantages?
OR
(b) Explain TCP socket programming and its applications?

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BCA CBCS-V Semester Regular Examinations NOV-2025

PAPER: OBJECT ORIENTED SYSTEMS DEVELOPMENT

Time: 3 Hours

Max Marks: 70

SECTION-A

I. Answer any FIVE of the following questions

(5×4=20Marks)

1. What is Analysis and Design? What is the main advantage of Object Oriented Development?
2. What is an Object? Give an example. Distinguish between Method and Message in Object.
3. What are Activity Diagrams? What is their purpose?
4. Define Events and Signals.
5. Differentiate Patterns and Frameworks.
6. What is the purpose of Deployment Diagrams? Name the basic elements of these diagrams.
7. What is the Role of UML in Unified Process?
8. What are the Four Types of Work Flows?

SECTION-B

II. Answer the following questions

(5×10=50Marks)

9. A) Explain Common mechanisms used in Basic Structural Modeling.

OR

- B) Explain about Building Blocks of UML.

- 10.A) Develop Use Case Diagram for Railway Reservation System.

OR

- B) What are State Machines? How to draw State Chart Diagram? Explain with an example.

- 11.A) What are Artifacts? Describe how to draw Artifact Diagrams?

OR

- B) Describe about Deployment and Collaborations.

- 12.A) Describe the Phases of Unified Process.

OR

- B) Explain the Advantages and Challenges of Architecture-Centric Process.

- 13.A) Describe Four Ps of Unified Software Development Process.

OR

- B) Compare and Contrast Incremental and Interactive Processes.

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BCA CBCS-V Semester Regular Examinations NOV-2025

PAPER: SOFTWARE QUALITY TESTING

Time: 3 Hours

Max Marks: 70

SECTION-A

I. Answer any FIVE of the following questions

(5x4=20Marks)

1. What is Software Quality? How you will measure it?
2. Specify the Core Ethical Principles in Software Quality.
3. Differentiate Quality Control and Quality Assurance.
4. What are Cyclomatic Complexity Function points?
5. How is Software Testing related to Software Quality?
6. What will be the content in Test Report?
7. Define Test Plan. Mention Test Plan Attachments.
8. Specify the Criteria for Test Completion.

SECTION-B

II. Answer the following questions

(5x10=50Marks)

9. A) Discuss about Software Processes and Methodologies.

OR

- B) Explain Total Quality Management Principles.

- 10.A) Discuss different methods in Analyzing Software Documents.

OR

- B) Explain various Software Quality Metrics.

- 11.A) Why Testing is an Engineering Activity? Explain.

OR

- B) What are the Strategies and Methods used for Black Box Test Case Design?

- 12.A) Explain the Components in Test Planning.

OR

- B) Discuss the Role of Three Groups in Test Policy Development.

- 13.A) Describe the Measurement and Milestones for controlling and monitoring Reports.

OR

- B) Discuss different Types of Reviews.

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BCA CBCS-V Semester Regular Examinations NOV-2025

PAPER: INTERNET PROTOCOLS

Time: 3 Hours

Max Marks: 70

SECTION-A

I. Answer any FIVE of the following questions

(5x4=20Marks)

1. Define networks and explain types of networks?
2. Explain about Fiber optics?
3. Explain CRC?
4. Write about classes in IP addressing ?
5. Write about IP4 versus IP6?
6. Explain BGP network types ?
7. Explain FTP and SMTP communication protocols?
8. Explain about Domain name spaces(DNS) ?

II. Answer the following questions

(5x10=50Marks)

9. (a) Explain network Topologies and types?
OR
(b) Explain TCP/IP model in network ?
10. (a) Explain Hamming code and advantages of Hamming code ?
OR
(b) Explain Stop and Wait ARQ and selective repeat ARQ ?
11. (a) Explain Distance Vector routing protocol ?
OR
(b) Explain ICMP protocol in network layer ?
12. (a) Explain about Multiplexing ?
OR
(b) Explain services of transport layer ?
13. (a) Explain about Command processing file transfer ?
OR
(b) Explain DNS in internet ?
